

Software Requirements Specification (SRS): Banking Management System

1. Project Overview

The Banking Management System is a comprehensive web-based platform designed to facilitate secure financial transactions, account management, and intelligent financial oversight. It bridges the gap between traditional retail banking and modern AI-driven financial management.

2. Executive Summary

The system provides three distinct user interfaces: **Customers** (managing personal finance), **Employees** (processing loan applications and administrative tasks), and **Admins** (system-wide oversight and monitoring). Key features include real-time transaction processing, automated fraud detection, AI-powered financial coaching, and robust security protocols.

3. Stakeholders

- **Customers:** Individual account holders.
- **Bank Staff (Employees):** Personnel responsible for loan verification and account support.
- **System Administrators:** Personnel responsible for maintenance, system health, and security.
- **Regulatory Bodies:** Entities requiring audit trails and compliance reports.

4. Target Users

- **Retail Customers:** Seeking banking services, budgeting tools, and loan options.
- **Operational Staff:** Verifying identity, approving credit, and managing user support tickets.
- **System Admin:** Ensuring platform security and managing user roles.

5. Functional Requirements

- **FR-01: Authentication:** Secure multi-factor authentication (MFA) for all user roles.
- **FR-02: Account Management:** Customers can open/close accounts and view balances.
- **FR-03: Transaction Processing:** Real-time internal and external fund transfers.
- **FR-04: Bill Payment:** Interface for scheduling and paying recurring/one-time bills.
- **FR-05: Loan Management:** Workflow for application submission, review, and approval/rejection.
- **FR-06: Fraud Detection:** Automated identification of suspicious transaction patterns.
- **FR-07: AI Financial Insights:** Engine to analyze spending habits and provide budgeting recommendations.
- **FR-08: Administrative Dashboard:** Tools for system monitoring and user management.

6. Non-Functional Requirements

- **Security:** Data encryption at rest and in transit; compliance with banking standards (e.g., PCI-DSS).
- **Availability:** 99.9% system uptime during business hours.
- **Scalability:** Ability to handle concurrent transactions as the user base grows.
- **Performance:** All transaction queries must resolve in under 2 seconds.
- **Auditability:** Immutable logs for every financial transaction and administrative action.

7. User Stories

- **As a Customer,** I want to receive spending insights so I can adjust my monthly budget.
- **As a Customer,** I want to apply for a loan online so I don't have to visit a physical branch.
- **As an Employee,** I want to view a queue of loan applications to perform timely approvals.
- **As an Admin,** I want to see a fraud alert log so I can secure the system against malicious activity.

8. Acceptance Criteria

- **Loan Application:** Loan must move from "Submitted" to "Pending" to "Approved/Rejected" based on Employee input.

- **Budgeting:** Budget recommendations must update at least once every 24 hours based on recent transactions.
- **Security:** Access is denied unless MFA is validated successfully.

9. Business Rules

- **Loan Limits:** Loans cannot exceed 5x the customer's verified monthly income.
- **Fraud Threshold:** Transactions exceeding \$10,000 from a new device trigger mandatory manual review.
- **Authentication:** MFA tokens expire after 5 minutes.

10. Assumptions

- The system integrates with a third-party KYC (Know Your Customer) provider for identity verification.
- Market data for budgeting insights is accessible via standard financial data APIs.

11. Constraints

- The application must support localization for different currency formats.
- Must be accessible via both desktop and mobile web browsers.

12. Risks

- **Data Breach:** Exposure of sensitive financial data (Mitigation: Encryption and periodic pen-testing).
- **Model Bias:** AI insights providing poor financial advice (Mitigation: Human-in-the-loop review for automated suggestions).
- **False Positives:** Legitimate transactions flagged as fraud (Mitigation: User override/confirmation flows).

13. Out of Scope

- Physical hardware (ATMs, kiosks).

- Trading or investment brokerage features.
- Crypto-currency wallet management.
- Physical branch interior management systems.

14. Open Questions

- Should the AI insight engine be integrated with external bank accounts (Open Banking)?
- What is the specific escalation process if the AI fraud detection flags a high-priority user incorrectly?
- Which specific compliance standards (e.g., GDPR, CCPA) must be strictly enforced?